Predicting the final grade of secondary-school students

Machine Learning × Advanced Python, final project, PGE 2

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1. Problem and goal

Who would use the tool. A school counsellor who wants to find students likely to finish the year with a low final grade, early enough to offer support.

Decision supported. Who should receive extra attention, and when.

Target and task. Regression on G3, the final grade, an integer from 0 to 20.

Cost of mistakes. Overpredicting a struggling student means missing someone who needs help, which is the more serious error. Underpredicting a student who is doing fine wastes some support time. This is why the main metric is RMSE, which penalises large errors, with MAE as a second, easier-to-explain metric.

Design choice. The dataset also contains G1 and G2, the grades of the first two periods of the same year. They are very predictive of G3, but they are only known late in the year. I excluded them on purpose, so the tool works early, before any grade exists. The price is a lower accuracy, which is reported honestly below.

2. Data and preparation

The data has 649 students and 30 features describing the student (age, sex, study time, past class failures, absences, wish to go to university, alcohol use), the family (parents' education and jobs, family size and relationships) and the school (school name, extra school support). It was loaded with the ucimlrepo package and is published under the CC BY 4.0 licence.

Checks. There are no missing values and no duplicate rows. The target ranges from 0 to 19.

Special code. 15 students (about 2%) have G3 = 0. This is probably a code for students who dropped out or skipped the final exam, not a real score: their average number of absences is 0, which suggests their absences were never recorded. I kept them, because a counsellor would meet such students in real life, and removing them would make the tool look better than it is.

Leakage. G1 and G2 are excluded (see above). All other features are known before the final exam.

Split. The data was split once, 80% for training (519 students) and 20% for testing (130 students), stratified on the quartiles of the target. The split uses a fixed seed (42) and the test set was not used for any choice.

Weak links. The strongest correlation of a numeric feature with G3 is with past failures (-0.39). Study time (+0.25), mother's education (+0.24), father's education (+0.21) and alcohol use (-0.20 and -0.18) follow. All links are weak to moderate, so a modest accuracy is expected without G1 and G2.

3. Method

Pipeline. All preparation lives inside a scikit-learn Pipeline with a ColumnTransformer. Numeric columns get median imputation and standardisation. Categorical columns get most-frequent imputation and one-hot encoding, and unknown categories are ignored at prediction time. Because the preparation is inside the pipeline, it is re-fitted on the training part of each cross-validation fold. If the data were scaled before the split, the validation folds would influence the mean and standard deviation, and the scores would be too optimistic (leakage).

Cross-validation. 5-fold cross-validation repeated 2 times, with the same folds and the fixed seed for every model. The metric is RMSE in grade points. The fold-to-fold standard deviation is about 0.3, so differences much smaller than that are not meaningful.

Models. A Dummy regressor that always predicts the average grade, and plain linear regression, serve as baselines. Four models were then tuned with GridSearchCV on the training set only:

- **Ridge** (alpha): linear regression with a penalty that shrinks the coefficients.
- **Lasso** (alpha): the same with a penalty that sets some coefficients to exactly zero.
- **k-NN** (number of neighbours, weights): predicts the average grade of the k most similar students.
- **Huber** (alpha, epsilon): a linear model whose loss is quadratic for small errors and linear for large ones, so a few outliers pull it less.

The Ridge grid was widened up to alpha = 1000 after the first run chose the edge of the grid, to check that the best value lies inside the grid.

Why a robust loss. With the squared-error loss, the 15 students with G3 = 0 produce about 31% of the total squared error of the Ridge model, although they are only 2% of the students. This is the reason for adding Huber loss as a fourth model.

4. Results

Table 1. Cross-validated RMSE on the training set (lower is better).

Model	Tuned hyperparameters	Best value	CV RMSE
Dummy (average grade)	none	none	3.19
Linear regression	none	none	2.76
Huber	alpha, epsilon	10, 1.35	2.71
Ridge	alpha	100	2.71
Lasso	alpha	0.05	2.73
k-NN	k, weights	20, distance	2.81

All models beat the Dummy baseline. Huber, Ridge and Lasso differ by less than 0.02, far below the fold-to-fold spread, so they are effectively tied, and k-NN is clearly weaker. I chose **Huber**, because it ties for the best score and its loss is the one that is justified by the outliers. Compared with the Dummy, its cross-validated RMSE is about 15% lower.

Out-of-fold check on the training set. For the chosen model, RMSE is 2.72 and MAE is 1.94. The 15 students with G3 = 0 account for 33% of the squared error, and the RMSE without them drops to 2.25. Huber loss therefore did not remove their effect, which is expected: no feature tells the model that a student will skip the exam.

Table 2. Final result on the test set (used once).

Metric	Cross-validation (train)	Test (130 students)
RMSE	2.72	**2.85**
MAE	1.94	**2.03**

The chosen model was refitted on the whole training set and scored once on the held-out students. The test error is slightly higher than the cross-validation error, but the gap is smaller than the fold-to-fold spread, so there is no sign of overfitting. On average, the prediction is off by about 2 grade points.

The residual plot shows most residuals between -4 and +4, scattered around zero without a clear curve. Three points lie far below zero (about -8, -9 and -12): they are overpredictions, most likely students with G3 = 0 whom the model expected to pass. The predictions only range from about 7 to 15, so the model stays close to the average and cannot identify extreme cases. Nothing was changed after seeing the test score.

5. Interpretation

The table lists the largest coefficients of the Huber model. Numeric features are standardised, so one unit means one standard deviation, and the yes/no pairs are one binary feature.

Feature	Coefficient	Meaning
Past failures	-0.75	Each extra standard deviation of failures lowers the predicted grade by about 0.75 points. This is the strongest effect.
Wants higher education (yes)	+0.55	Students who want to continue to university are predicted about 1.1 points higher than those who do not.
Extra school support (yes)	-0.57	Students who receive it are predicted lower.
Father's job (teacher)	+0.46	A teacher father goes with a higher prediction than a services job (-0.47).
School (GP)	+0.43	School GP is predicted higher than school MS (-0.43).
Study time	+0.31	More weekly study time raises the prediction.
Absences	-0.29	More absences lower the prediction.

These are associations, not causes. Extra school support probably does not lower grades: weaker students are the ones who receive it. The coefficients are also shrunk by regularisation (alpha = 10), so they are conservative. A student with three past failures instead of none gets a prediction about 3.7 points lower in the command-line tool, which is large because most students have no failures, so three failures is many standard deviations away.

6. Limits of the model

1 **It cannot find the students who matter most.** Without G1 and G2, predictions stay between about 7 and 15, and the students with G3 = 0 (dropout or skipped exam) are not detected. They produce the largest errors.

2 **Small and narrow data.** There are only 649 students from two Portuguese schools. The average error is about 2 grade points, the differences between the best models are within the noise of cross-validation, and the model may not transfer to other schools or countries.

7. Risks of using it on real people

A wrong prediction can label a student unfairly: a low prediction may reduce expectations or attention, and a high one may hide a student who needs help. Several features act as proxies for social background (school, father's job, wish for higher education), so the model could reproduce existing inequalities and treat students differently because of their family rather than their effort. The data also contains sensitive personal
information (family situation, alcohol use, health, romantic relationship), which raises privacy concerns about storage, access and consent. The tool should only support a human decision by the counsellor and never replace it, and students should not be ranked or sanctioned from its output alone.

8. Reproducibility

The work is a Python package with a src/ layout: data.py (class DatasetLoader: download, clean, split), features.py (the ColumnTransformer), models.py (class Trainer: pipelines, cross-validation, comparison, save and load), evaluate.py (metrics) and cli.py. A config.yaml file holds the seed, the split size, the cross-validation settings and the hyperparameter grids, so no path or parameter is hard-coded. Versions are pinned in pyproject.toml. 14 pytest tests cover the cleaning, the split (no test row in train), the pipeline (the scaler is fitted on train only, unseen categories are handled), the baseline, the range of predictions (between 0 and 20), saving and loading, and the command-line tool. The code is checked with ruff. The command-line tool has three commands: student-grades train, student-grades evaluate and student-grades predict --input examples/new_case.json. The notebook calls the package, and the package reproduces the notebook's test result (RMSE 2.85, MAE 2.03). The README explains the installation and usage, and AI_USAGE.md lists the use of generative AI.

References

- P. Cortez, *Student Performance*, UCI Machine Learning Repository, https://archive.ics.uci.edu/dataset/320 (CC BY 4.0).
- P. Cortez and A. Silva, *Using Data Mining to Predict Secondary School Student Performance*, FUBUTEC 2008.
- scikit-learn documentation, https://scikit-learn.org.